

Taka Khoo

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takahoo.com |  | 

EXPERIENCE

Mozart AI (\$7M+ raised; Balderton-backed)

Mar 2026–Aug 2026

Founding Engineer, AI & Product, London

- Shipped TypeScript/React library, onboarding, and authenticated Liveblocks collaboration for a platform with **250K+ users** and **1M songs generated**; persisted track/mixer/MIDI state.
- Built multi-model audio evaluation: BPM/key/lyric checks, stem/semantic metrics, Bradley–Terry A/B ranking, PostHog telemetry, and a LiveKit voice-agent pipeline.

DARPA INGOTS (NARF Industries & Dartmouth LISP Lab)

Jun 2025–Jun 2026

AI Specialist / Cybersecurity Researcher

- Built AI-assisted Android vulnerability discovery, exploitability triage, verifier-gated PoCs, and structured evidence; developed cross-privilege exploit chains using Ghidra, ADB, and SELinux.
- Extended ParseRE's QEMU binary labeling with an ELF corpus and grammar; built libelf/json-c/curl Docker harnesses and reproducible Cuttlefish evaluations.

Dartmouth Thayer School of Engineering

Jun 2025–Jun 2026

Head Teaching Assistant, Graduate Machine Learning

- Led two graduate ML courses; wrote [four textbooks](#) expanding Prof. Peter Chin's lectures, now used at Dartmouth.

Igloo Music

Feb–Jun 2024

Production & Mastering Engineer, Los Angeles

- Patched mic/preamp/synth signal chains and plug-ins; supported production, mastering, and [IMDb-credited dubbing](#) in a Grammy-winning studio.

SELECTED PRODUCTS, RESEARCH & THESES

MODULO, AI-Native Music Workstation

Independent product; M.S. thesis

- Built a C++/JUCE/Tracktion workstation with Python ML, VST/AU hosting, real-time audio, generative chords, audio-to-MIDI, and stem separation.
- [First-authored the revised UIST manuscript](#): a within-subjects study halved time to satisfaction, tripled chord roots, nearly quadrupled chord qualities, and earned unanimous preference.

Token U-Net, Neural Audio Restoration

Engineering honors thesis

- Built a 1.08B-parameter EnCodec-token U-Net with FiLM and five-stage curriculum learning to restore full-mix music degraded by EQ, gain, compression, reverb, and echo.

My Life in Sound, Thesis Album

Music honors thesis

- Composed, performed, recorded, mixed, and mastered a multi-genre thesis album across five instruments.

Federated Speech Privacy

First-author research; code

- Reconstructed speech features from federated ASR gradients under CTC loss; evaluated inversion on LibriSpeech.

EDUCATION

Dartmouth College

Hanover, NH

M.S. Engineering Sciences, AI / Signal Processing / Music (*GPA: 3.92/4.00*)

Jun 2026

A.B., Engineering Sciences & Music, honors in both majors

Jun 2025

SAT: 1590/1600 (800 Math, 790 Verbal)

SKILLS & INTERESTS

Engineering & AI: C++, Python, TypeScript/JavaScript, React, Node.js, SQL/PostgreSQL, GCP, Docker, Git; PyTorch, TensorFlow, LLM agents, **Model Context Protocol (MCP)**, RAG, AI evaluation, audio/speech ML, DSP.

Product engineering: **0-to-1 full-stack systems**, real-time collaboration, API/data architecture, product analytics, A/B evaluation, rapid prototyping; Liveblocks, LiveKit, PostHog, JUCE, Tracktion Engine; **DAWs (expert)**.

Leadership: Co-founded Jump Ahead Tutoring (50+ tutors); coached 10+ Ivy League admits. **NCAA Division I swimmer**; broke Dartmouth school records twice and Hawaiian state records; perfect-pitch multi-instrumentalist.